PAGE 10 – BENCHMARKING WORKSHEET

Enter the empirical execution results recorded by your benchmarking module. Do not use theoretical guesses.

BENCHMARK RESULTS TABLE

Algorithm	Dataset Size	Comparisons	Swaps	Execution Time	Theoretical Big-O
Bubble Sort	Small (N=10)	35	19	0.001	O(n²)
Bubble Sort	Medium (N=500)	124120	65743	0.849	O(n²)
Bubble Sort	Large (N=5000)	12491395	6251385	74.611	O(n²)

Algorithm	Dataset Size	Comparisons	Swaps	Execution Time	Theoretical Big-O
Selection Sort	Small (N=10)	45	7	0.001	O(n²)
Selection Sort	Medium (N=500)	124750	495	0.368	O(n²)
Selection Sort	Large (N=5000)	12497500	4993	28.684	O(n²)

EMPIRICAL ANALYSIS QUESTIONS

1. **Did execution time scale as expected when dataset size increased?**

Explain: Mostly yes. Going from N=500 to N=5000 multiplies N by 10, so an O(n²) sort should do about 100 times more work. Bubble Sort's comparisons grew from 124,120 to 12,491,395 (about 100×), and Selection Sort's from 124,750 to 12,497,500 (about 100×). Execution time grew by a similar amount. Bubble Sort rose from 0.849 ms to 74.611 ms (about 88×) and Selection Sort from 0.368 ms to 28.684 ms (about 78×). Both are close to the predicted 100×, but a bit under it. At N=500, the run takes under a millisecond, so fixed overhead and measurement noise make up a larger share of the time. At N=10, the 0.001 ms times are at the limit of the timer's resolution, so they can't be compared meaningfully.

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2. **Did your empirical results match theoretical Big-O predictions?**

Explain: Yes. Both sorts are O(n^2), and the comparison counts follow it closely. Selection Sort's comparisons exactly equal $n(n-1)/2$: 45 at N=10, 124,750 at N=500 and 12,497,500 at N=5000. Bubble Sort came in slightly lower (124,120 and 12,491,395) because it can stop early once a pass has no swaps. The swap counts show a difference that Big-O doesn't capture. Bubble Sort made 6,251,385 swaps at N=5000, while Selection Sort made only 4,993, which grows linearly. Both algorithms are still O(n^2) because their comparison counts dominate, but swap cost explains why Selection Sort was about 2.6 times faster at N=5000 (28.684 ms against 74.611 ms).

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3. **What hardware or execution factors cause actual execution time to differ from theoretical Big-O?**

Actual execution time differs from theoretical Big-O because of hardware and execution factors such as CPU cache and memory access, branch prediction, compiler optimizations, operating system activity, and changes in CPU frequency or temperature. Timer resolution and measurement noise can also affect results, especially for very short runs. Additionally, algorithms with the same Big-O complexity may perform differently because of constant factors, such as the number of comparisons and swaps they perform.